Application of Information & Communication Technologies

Lecture-19

Recap of Lecture 18

- Types of Databases
- Functions of DBMS
- DBMS Software Examples (MS Access, MySQL)
- Role of Databases in ICT Applications

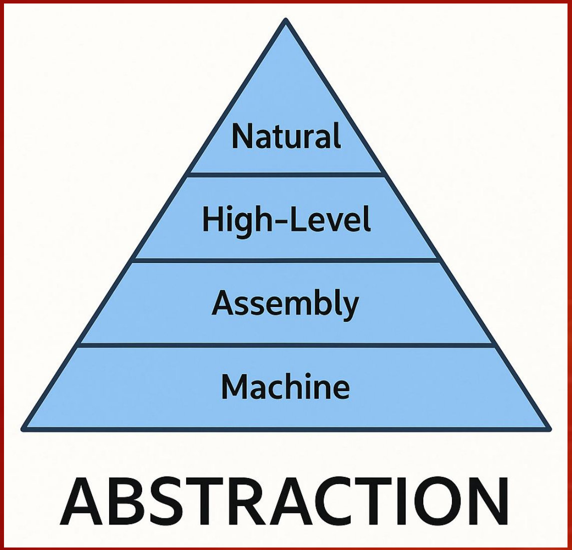
Overview of Lecture 19

- Introduction to Programming Languages
- Learning Objective:
  - Understand what programming languages are, their types, and how different languages express the same logic.

Programming Language

- A programming language is a way to communicate with the computer.
- It gives instructions for performing specific tasks.
- Used to develop software, websites, apps, and games.
- 🖥️ Example:
  - Human → Language → English
  - Computer → Language → C++, Python, Java

Types of Programming Languages



Category	Description	Examples
Low-Level	Machine-oriented (close to hardware)	Assembly, Machine Code
High-Level	User-friendly, easy to read/write	C, C++, Java, Python
Very High-Level	Abstract, used in automation & AI	MATLAB, SQL, Scratch

Translators in Programming

Translator Type	Function	Example Languages
Assembler	Converts assembly code → machine code	Assembly
Compiler	Converts entire code at once → executable	C, C++
Interpreter	Translates code line-by-line	Python, JavaScript

- Compiler = Fast Execution
- Interpreter = Easy Debugging

Hello World programs

C++

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << "Hello, World!";
    return 0;
}
```

Java

```java
class Main {
  public static void main(String[] args) {
    System.out.println("Hello, World!");
  }
}
```

Python

```python
print("Hello, World!")
```

C#

```csharp
using System;

class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("Hello, World!");
    }
}
```

Comparing Languages

Language	Code Length	Type	Execution Speed	Ease of Learning
C++	Long	Compiled	Fast	Moderate
Python	Short	Interpreted	Moderate	Easy
Java	Medium	Hybrid	Fast	Moderate

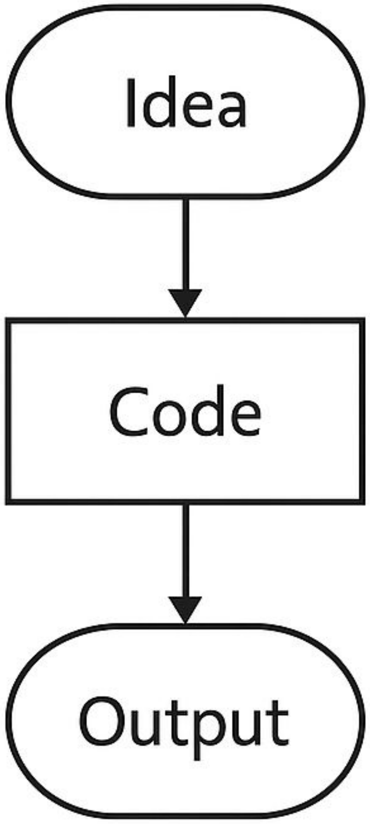
Why Learn Programming?

Turns ideas into working software.

Improves problem-solving skills.

Needed for data science, AI, web, and app development

Builds logic and creativity



Summary

- Programming = communicating with the computer
- Types: Low-level, High-level, Very High-level
- Translators: Assembler, Compiler, Interpreter
- Example languages: C++, Java, Python
- All languages solve the same problems in different ways

Suggested Reading

- Ch-13, The System Unit: Processing and Memory , “Understanding Computers: Today and Tomorrow, Comprehensive”, 15th Edition by Deborah Morley & Charles S. Parker